

ZHOU, CONGHAO

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RESEARCH INTERESTS

dark matter · gravitational lensing · galaxy clusters · large scale structure
shear calibration · image simulation

EDUCATION

	University of California, Santa Cruz
Expected 2027	Ph.D. in Physics Physics Department Thesis: <i>TBD</i> Committee: TESLA E. JELTEMA, ALEXIE S. LEAUTHAUD
	Duke University
2020	B.A. in Physics with Highest Distinction Department of Physics Thesis: <i>Intrinsic Alignment of redMaPPer Clusters in the Dark Energy Survey</i> Committee: MICHAEL A. TROXEL, CHRISTOPHER W. WALTER, ROBERT L. WOLPERT

SCIENCE PUBLICATIONS WITH MAJOR CONTRIBUTIONS

[†] STUDENT ADVISED

1. No Cluster Left Behind I: Joint Mass Calibration for Dark Energy Survey and South Pole Telescope Clusters, in prep.
C. ZHOU, A.N. SALCEDO, H.-Y. WU, S. GRANDIS et al.
2. Response Bias in Shear Calibration, in prep.
C. ZHOU, M.R. BECKER, M. GATTI et al.
3. Cluster finding with outskirts stellar masses and percolation
[arXiv:2608.19768 \(2026\)](#)
P. AVALOS[†], C. ZHOU, T. JELTEMA, K. LEIDIG, S. XU et al.
4. Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal
[arXiv:2606.21141 \(2026\)](#)
C. ZHOU, X. LI et al.
5. Relationship between 2D and 3D galaxy stellar mass and correlations with halo mass
[J. Cosmol. Astropart. Phys. 10, 014 \(2025\) \[arXiv:2502.05158\]](#)
C. ZHOU, A. LEAUTHAUD, S. XU, B. DIEMER, S. HUANG et al.
6. Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data
[Phys. Rev. D 110, 103508 \(2024\) \[arXiv:2312.11789\]](#), *Editors' Suggestion*
C. ZHOU, H.-Y. WU, A.N. SALCEDO, S. GRANDIS, T. JELTEMA, A. LEAUTHAUD et al.
7. The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters
[Mon. Not. R. Astron. Soc. 526, 323 \(2023\) \[arXiv:2302.12325\]](#)
C. ZHOU, A. TONG, M.A. TROXEL, J. BLAZEK, C. LIN et al.

TECHNICAL PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Surface brightness profiles around massive galaxies in LSSTComCam data
[LSST Technote SITCOMTN-165 \(2025\)](#)
C. ZHOU, T. JELTEMA, A. VON DER LINDEN, L.S. KELVIN et al.

CONTRIBUTING AUTHOR SCIENCE PUBLICATIONS

1. The Connection between Halo Assembly History and the Stellar Distribution I: The Central Stellar Mass–Richness Plane
[arXiv:2609.14488 \(2026\)](#)
S. XU et al.
2. Probing Projection Effects in Optically-Selected Clusters with Velocity Dispersions Outliers
[arXiv:2609.08095 \(2026\)](#)
K. GRAHAM et al.
3. OASIS: Observation-Aware Simulation-Based Inference via Distributional Matching
[arXiv:2606.22572 \(2026\)](#)
A. FARAHİ et al.
4. The Vera C. Rubin Observatory Data Preview 1
Astron. J. **171**, 360 (2026) [[arXiv:2603.23786](#)]
VERA C. RUBIN OBSERVATORY TEAM et al.
5. DESI-DR1 3×2 -pt analysis: consistent cosmology across weak lensing surveys
Open J. Astrophys. **9** (2026) [[arXiv:2512.15960](#)]
A. PORREDON et al.
6. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)
Astrophys. J. Lett. **1001**, L35 (2026) [[arXiv:2507.13409](#)]
C.O. CHANDLER et al.
7. Data Release 1 of the Dark Energy Spectroscopic Instrument
Astron. J. **171**, 285 (2026) [[arXiv:2503.14745](#)]
DESI COLLABORATION et al.
8. Impact of projection-induced optical selection bias on the weak lensing mass calibration of galaxy clusters
Phys. Rev. D **113**, 063559 (2026) [[arXiv:2510.00753](#)]
T.N. NDE et al.
9. Reaching for the Edge II: Stellar Halos out to Large Radii as a Tracer of Dark Matter Halo Mass
[arXiv:2511.10723 \(2025\)](#)
K. LEIDIG et al.
10. Cosmological Constraints from Dark Energy Survey Year 1 Cluster Lensing and Abundances with Simulation-based Forward-Modeling
[arXiv:2510.25706 \(2025\)](#)
A.N. SALCEDO et al.
11. Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics
[arXiv:2510.17996 \(2025\)](#)
Y. LUO et al.
12. Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering
Phys. Rev. D **112**, 083535 (2025) [[arXiv:2503.13632](#)]
T.M.C. ABBOTT et al.
13. The Outskirt Stellar Mass of Low-redshift Massive Galaxies Is an Excellent Halo Mass Proxy in Illustris/IllustrisTNG Simulations
Astrophys. J. **990**, 127 (2025) [[arXiv:2412.03406](#)]
S. XU et al.
14. Association between optically identified galaxy clusters and the underlying dark matter halos
Phys. Rev. D **112**, 043517 (2025) [[arXiv:2506.17526](#)]
S. CAO et al.

15. Improving galaxy cluster selection with the outskirts stellar mass of galaxies
[Phys. Rev. D 111, 123524 \(2025\)](#) [[arXiv:2410.20205](#)]
M. KWIECIEN et al.
16. Optical galaxy cluster mock catalogs with realistic projection effects:
Validations with the SDSS clusters
[Phys. Rev. D 111, 063502 \(2025\)](#) [[arXiv:2410.02497](#)]
A. LEE et al.
17. Dark Energy Survey Year 3 results: Simulation-based cosmological inference
with wavelet harmonics, scattering transforms, and moments of weak lensing
mass maps. Validation on simulations
[Phys. Rev. D 109, 063534 \(2024\)](#) [[arXiv:2310.17557](#)]
M. GATTI et al.
18. Building an Efficient Cluster Cosmology Software Package for Modeling
Cluster Counts and Lensing
[arXiv:2309.06593 \(2023\)](#)
M. AGUENA et al.
19. Dark Energy Survey Year 3 Results: Deep Field optical + near-infrared images
and catalogue
[Mon. Not. R. Astron. Soc. 509, 3547 \(2022\)](#) [[arXiv:2012.12824](#)]
W.G. HARTLEY et al.

FUNDING

2026–2027	Chancellor’s Dissertation Year Fellowship , UC Santa Cruz, \$33,000
2020	Rodney I. McCormick Award , Duke University, \$1,500

TALKS

aug. 2026	Tsinghua University , Department of Astronomy Seminar <i>No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data</i>
jul. 2026	Tsung-Dao Lee Institute , Astronomy Seminar <i>No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data</i>
jul. 2026	Shanghai Jiao Tong University , Department of Astronomy Seminar <i>No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data</i>
may 2026	South Pole Telescope Collaboration , Cluster Working Group Meeting <i>No Cluster Left Behind: Joint Mass Calibration for DES and SPT Clusters</i>
dec. 2024	University of Arizona , Theoretical Astrophysics and Cosmology Seminar <i>Cluster Cosmology: Pixel to Object, CMB to Stellar Mass</i>
nov. 2024	University of California, Davis , Cosmology Seminar <i>Optical Cluster Cosmology: Selection to IA, Pixel to Object</i>
apr. 2024	CMB-S4 Collaboration , Cluster AWG Meeting <i>Calibrating Optical Selection Bias with Multiwavelength Data</i>
mar. 2024	Brown University , Center for the Fundamental Physics of the Universe Seminar <i>Cluster Cosmology Beyond Richness</i>
mar. 2024	University of Pennsylvania , Journal Club <i>Response Clustering Bias</i>
oct. 2023	Tsinghua University , Cosmology Seminar <i>Intrinsic Alignment in DES Y1 redMaPPer Clusters</i>
oct. 2023	Stony Brook University & Brookhaven National Lab , Cosmology Seminar <i>Constraining Optical Selection Bias in Cluster Lensing w/ Multiwavelength Data</i>

OBSERVING EXPERIENCE

<i>10.5 effective nights</i>	Dark Energy Spectroscopic Instrument, Mayall 4m at Kitt Peak
<i>3 nights</i>	DECaLS Intermediate Band Imaging Survey, Blanco 4m at CTIO

SCIENCE COLLABORATIONS

<i>2026–present</i>	Roman Science Collaboration
<i>2024–present</i>	Rubin Observatory Commissioning, In-Kind Contributor
<i>2023–present</i>	LSST Galaxies Science Collaboration
<i>2023–present</i>	LSST Informatics & Statistics Science Collaboration
<i>2020–present</i>	LSST Dark Energy Science Collaboration, Full Member since 2024
<i>2020–present</i>	Dark Energy Spectroscopic Instrument, Sponsored Member
<i>2018–present</i>	Dark Energy Survey, Student Member

PROFESSIONAL SOCIETIES

<i>2026–present</i>	Sigma Xi, Member
<i>2024–present</i>	American Astronomical Society, Student Member
<i>2024–present</i>	American Physical Society, Student Member

OUTREACH

<i>2023–2024</i>	Starlight Over Street Light
<i>2022</i>	UC Santa Cruz Physics Outreach Taskforce
<i>2022</i>	Science Internship Program Mentor
<i>2022</i>	UC Santa Cruz Physics Mentoring Program Mentor

COLLABORATION SERVICE

<i>jun. 2026–present</i>	LSST DESC Speaker's Bureau
<i>jul. 2022–present</i>	LSST DESC Mentoring Committee
<i>apr.2022–may.2025</i>	DES Early Career Scientists Committee

SEMINAR & MEETING ORGANIZATION

<i>2025 Winter</i>	UC Santa Cruz Physics Colloquium Committee
<i>sep.2023–jun.2024</i>	Organizer, CosmoGal Seminar, UC Santa Cruz
<i>jul.2023–jun.2024</i>	Organizer, BCG Seminar, Boise State
<i>jan. 2023</i>	LOC, APS Conference for Undergraduate Women in Physics at UC Santa Cruz
<i>aug. 2022</i>	SOC, LSST DESC 2022 Chicago Meeting

REFEREEING

2024–present Astronomy & Astrophysics

TEACHING EXPERIENCE

2024 Spring Application of Quantum Materials (Grader)

2024 Winter Electricity, Magnetism, and Optics (TA)

2022 Fall Quantum Mechanics (TA)

2022 Summer Intermediate Physics Laboratory (TA)

2022 Spring Advanced Astrophysics Laboratory (TA)

3 times Introductory Physics Laboratory (TA)

September 25, 2026